

BrickAtlas: Separating Visual Binding and Compositional Reasoning in Source-Grounded Repair Problems

Supplementary Material

Anonymous CVPR submission

Paper ID Draft

1. Version and Evidence Boundary

The primary version is `visual-repair-v1`, with one task family: visually bound minimum terminal repair. Its release contains 84 constructions, 1008 repair observations and 15 dependence groups. Twelve views of a construction are not twelve independent tasks. The main paper’s figures and result schemas refer to this version.

Earlier artifacts remain available for inspection and comparison: the `display-v3` atomic study has 73 Color pairs, 67 Part-type pairs and 67 position panels; its supporting hypothetical graph study has 219 distinct observations. The historical dossier has 617 items. The later CX1–CX3 illustrations concern symbolic repair, diagnosis and registration. These counts and outcomes are not pooled with the present family. In particular, a solved symbolic example does not establish visual necessity.

No human judgments or learned-model results are claimed for the primary version in this workspace. The result tables are deliberately blank. Deterministic construction checks, algorithmic baselines and synthetic scorer tests are real executable evidence, labeled as such. They do not substitute for collection.

2. Complete Task Contract

2.1. Native input and permitted evidence

Each input has a fixed question, an explicit undirected graph, unavailable vertex IDs, two fixed terminals, four candidate IDs, a budget and an output schema. Native images show an unlabeled reference and four labeled candidates with actual source geometry. The reference uses a different view. Public graph IDs are independent of historical part numbers and source-instance numbering.

Model-facing packets omit source name, construction and split IDs, intervention labels, source coordinates, private display mappings and GT. Such fields remain in evaluator indexes to preserve provenance and paired analysis. Source meshes and their original transforms are not extra

inputs to the visual condition.

The four conditions have different information boundaries:

Table 1. Condition access. A named correct terminal is privileged binding information. Atomic binding receives no structural problem.

Condition	Image	Graph/faults	True terminal
Multimodal	yes	yes	no
No-image	no	yes	no
Oracle binding	no	yes	yes
Atomic binding	yes	no	no

The exact representative question below is copied from a native packet, not paraphrased for publication.

Match the unlabeled reference geometry to exactly one candidate card; ignore orientation and use its current printed ID as `bound_terminal`. The graph IDs refer to the current display, not source part identities. Together with `fixed_terminals` it must lie in one connected component. Only paths inside the supplied undirected graph count. Initially every listed missing vertex and its incident edges is unavailable. Restoring one listed vertex costs 1 and reinstates its original edges to present neighbors. Return every minimum-cost restoration set, its own final component sizes, `minimum_cost`, and whether budget suffices. Graph unavailability is not a physical removal. Order is irrelevant; duplicate vertices/solutions and unlisted actions are invalid.

The source IDs, fault sets, public graph adjacency and evaluator answer for that example are recorded in `paper/repair-figure-provenance.json`. Complete packets and all other inputs are shipped with the version.

2.2. Answer semantics

The repair output is a JSON object containing a bound terminal, minimum cost, budget decision and an array of solu-

tions. Each solution contains a restoration set and component sizes. The solution array must contain every optimal restoration, exactly once. Atomic binding instead returns only the bound terminal.

```
{
  "bound_terminal": "V9801",
  "minimum_cost": 4,
  "budget_sufficient": true,
  "solutions": [
    {
      "restored": [
        "V3892",
        "V4504",
        "V5246",
        "V6716"
      ],
      "component_sizes": [
        11
      ]
    },
    {
      "restored": [
        "V3892",
        "V4504",
        "V6716",
        "V7052"
      ],
      "component_sizes": [
        11
      ]
    }
  ]
}
```

The scorer treats order within restoration sets, solution arrays and component-size multisets as irrelevant. It does not treat a duplicate ID as repeated restoration or silently discard an invalid extra solution. Unknown vertices, wrong types, nonfinite values and malformed structure are failures according to the version's schema. No result is inferred from a model's prose rationale.

If a model supplies a correct optimum for the wrong terminal, its structural calculation does not rescue exact task correctness. Binding and structural components are reported separately so this failure can be inspected. Conversely, a correct terminal with a feasible but nonminimal repair is not exact success.

2.3. Factorial and equivariance relationships

One construction has two fault states, each with anchor, visual change and visual preservation. Two independent ID replicates yield twelve repair observations. Fault states exchange one unavailable vertex with a present one; graph edges remain the recognized source edges. A changed terminal is selected by swapping the matching candidate's la-

bel with one distractor. Preservation swaps two distractor labels. Both edits touch two label regions, with all part pixels unchanged.

Admissibility checks complete optimal-solution sets, not one arbitrarily chosen solution. Every visual-changing comparison must alter those sets, every declared structural comparison must alter them with the image fixed, and every preserving comparison must keep the complete output invariant. ID replicates are decoded through private mappings before semantic comparison. Renaming an ID is not a new physical part or topology.

3. Sampling and Dependence

3.1. Population and exclusion ledger

The source population is the finite collection of 24 original assemblies containing 15,334 placed LDraw parts. The primary constructor enumerates complete non-root subassemblies including descendants, with 12–100 parts, connected recognized graphs and recorded connector coverage. It excludes candidates with a recorded unmatched intersection inside the selected induced set. Unrecorded contacts or intersections can still exist; this screening does not certify physical buildability.

The sampling ledger records every candidate, source/contributor identity, coverage screen, design signature, graph-template signature, overlap exclusion and structural search outcome. A content-seeded ordering ensures reproducibility. Within each source, retained instance sets do not overlap. Repeated aligned typed-distance designs are removed; potential topological duplicates are conservatively grouped.

3.2. Development and held-out membership

Sources linked by contributor, aligned design or graph-template signature are combined transitively. Development/held-out assignment is fixed at group level before bounded structural search. Both ID replicates and every intervention remain in the same construction and split. The held-out set tests construction groups from this declared finite population, not arbitrary unseen mechanisms.

Search attempts are capped and logged. The first proposal satisfying the admissibility contract is selected; no learned-model outcome or human answer is used to choose it. This produces a challenge distribution conditioned on informative counterfactuals. It is not an estimate of naturally occurring faults. Search failure and exclusion distributions are therefore part of the release.

3.3. Precision and sensitivity

Repeated views improve within-construction diagnosis without increasing the number of independent groups. For

Table 2. Exact complete adjacency of the main-paper development example. Source edges are undirected.

Vertex	Neighbors
V1470	V6716
V2353	V4504, V9801
V2647	V4504, V6716
V3673	V9801
V3892	V5246, V5885, V7052
V4504	V2353, V2647, V7372, V8726
V5246	V3892, V6716
V5885	V3892
V6716	V1470, V2647, V5246, V7052, V7372, V9308
V7052	V3892, V6716
V7372	V4504, V6716
V8726	V4504, V9801
V9308	V6716
V9801	V2353, V3673, V8726

Table 3. Primary construction coverage generated from the actual selection ledger. Sources or groups with no admitted construction do not reappear as additional independent observations.

Source	Split	Dependence group	Constructions	Observations
10128	dev	296d07d320de	3	36
10159	heldout	8d40d2fdacfd	1	12
10220	heldout	3a415d263677	9	108
31028	heldout	4fbded52c2a0	1	12
5867	heldout	d63cd142acd2	1	12
omr-10242	heldout	7701215dc34d	5	60
omr-10265	dev	2d8ad5396040	4	48
omr-10269	dev	0cc79961cf45	15	180
omr-21309	heldout	1ed5b08f321b	11	132
omr-31031	heldout	1bdc8c05fa6e	1	12
omr-31088	dev	a5d3ae06f426	2	24
omr-42004	heldout	5ec9ed82a3a3	2	24
omr-42020	dev	53fb9a1e6860	3	36
omr-42054	heldout	de06c1956435	12	144
omr-42061	heldout	de06c1956435	3	36
omr-42066	heldout	de06c1956435	10	120
omr-42102	heldout	621fdcee6fc0	1	12

a bounded independent group mean, a worst-case normal 95% half-width target of .20 gives $\lceil (1.96/.40)^2 \rceil = 25$ groups before any small-sample correction. This is a sensitivity target, not a claim that the available resource achieves that precision. Actual eligible group counts accompany every score and leave-one-group-out result.

Equal-group means first average within each dependence group and then across eligible groups. Micro means weight all planned observations. Paired comparisons use the same groups in both conditions. Group sensitivity does not establish a superpopulation sampling distribution, and no collection of random ID permutations repairs insufficient construction diversity.

4. Scoring, Certificates and Tests

4.1. Independent repair oracle

The constructor obtains certificates with graph search over restoration subsets. The evaluator independently rebuilds connected components using union-find. It enumerates candidates from public graph/fault records and the evaluator’s bound-terminal key, validating every minimum set and component output. A precomputed string is not used as the semantic criterion.

For binary exact outcomes c_{rsav} , visual-change Both and visual-preserving Both require both paired endpoints correct. Structural Both requires both fault states correct. Factorial All-six requires all six cells within one ID replicate

correct. The second replicate tests equivariance and remains dependent on the first. A stable incorrect answer receives no preservation credit.

Binding accuracy, repair given correct binding, exact repair, invalid rate, missing rate and received count are all separate fields. Conditional metrics with no eligible observations remain undefined. Missing, refused or malformed planned calls stay in the denominator. When a qualified subset is used, full-set outcomes and exclusions remain available rather than being overwritten.

4.2. Task-aligned joint outcomes

For task i , define binary outcomes A_i (atomic binding), O_i (oracle exact repair), M_i (multimodal exact repair) and B_i (multimodal binding). Alignment uses the exact task identity, including construction, fault, visual arm and ID replicate. Repeated atomic images remain aligned to each planned task; they do not supply additional independent evidence. The eight contingency cells are

$$n_{aom} = \sum_i [A_i = a, O_i = o, M_i = m], \quad (a, o, m) \in \{0, 1\}^3.$$

The joint failure numerator is n_{110} and the eligible denominator is $n_{110} + n_{111}$. We report their ratio as micro J , as well as the mean of within-group ratios over groups with nonzero eligibility. Counts, eligible groups and excluded zero-eligibility groups accompany both. An empty eligible set produces null estimates, not perfect performance.

For the same planned task set, the paired exact-repair contrast is the average of $M_i - O_i$ and the binding contrast the average of $B_i - A_i$. They retain their signs. Equal-group contrasts first average these differences within each group; micro contrasts average over observations. They are matched observed differences, not causal effects of disabling an internal model operation.

Marginal condition scores do not determine the joint diagnostic. A synthetic regression fixture has two groups with two constructions each and twelve cells per construction. Multimodal success occurs on the first construction in each group. Placing both isolated successes on the first versus second construction leaves every condition's accuracy at .5, including its group and intervention metrics, but changes joint failure from 0/24 to 24/24. These values are synthetic tests of the analysis, not model results. No product of marginal accuracies is used as an eligibility estimate.

For each comparison, subset and split, 2,000 draws with seed 20260922 resample the sorted complete dependence groups with replacement. The same draw indexes the joint numerator, eligible denominator and both contrast vectors. Group intervals recompute the mean of defined group values; micro intervals recompute the pooled ratio. Groups with zero joint eligibility remain in the draws. Draws containing no eligible group are undefined and omitted

from joint percentiles, with the defined-draw count reported. Joint intervals require at least two eligible groups; contrast intervals require at least two planned groups. Leave-one-group-out estimates preserve null empty cases. These are finite-resource sensitivity summaries, not claims of population coverage.

The run lock explicitly declares which atomic, oracle and multimodal configurations form one comparison. System identity, evidence kind, revision, settings and attempt count must match; the locked adapter and exact native input hashes apply throughout. Separately locked runs are not silently pooled. An absent receipt in a planned condition scores false. A condition not planned for that model makes the comparison unavailable; incomplete or mismatched planned task membership is rejected. The current protocol uses one attempt.

4.3. Adversarial and equivalence checks

Synthetic fixtures exercise alternative valid serialization and solutions, omitted or duplicate optima, nonminimal feasible sets, wrong terminals, unknown IDs, wrong component outcomes and missing or off-version receipts. Tests never populate empirical tables. The full test report accompanies the released evaluator. Exact-input hashes distinguish a genuine result for the selected observation from a response to a different input or condition. Additional fixtures check joint outcomes, absent conditions, incompatible locks, empty eligibility, construction-wide exclusion and equality of full and qualified estimates when all constructions qualify. Adapter failures are exercised with mocked subprocesses.

4.4. Legal baselines and restricted interpretation

The local no-purchase study includes only explicitly defined algorithmic configurations. Legal baselines use current public pixels and text. Coordinate or private mapping access is oracle information and cannot be silently included in those baselines. Pixel shape/appearance matchers, card-position rules, sorted/numeric ID rules and graph heuristics probe different shortcuts.

A modular pixel binding system followed by exact graph optimization can legitimately solve the full public task. Its success is reported, not treated as forbidden access. If a legal shallow baseline exceeds .80 held-out binding or full-repair accuracy, the interpretation must explicitly describe a shallow-solvable binding stage. Selection is not repeated to conceal that result.

Shallow-solvable binding stage: no strong perceptual-difficulty claim.

5. Human Qualification and Run Lock

Human qualification concerns exact rendered evidence, while the optimization certificate concerns the supplied

Table 4. Actual deterministic baseline audit. These are local algorithm outcomes; they are separate from the blank learned-model tables. Full per-condition denominators and configurations are retained in the versioned baseline report.

Algorithm	Condition	Bind	Exact	All-six
text_first-repair	no_image	21.0	21.0	0.0
min_id-repair	no_image	20.1	20.1	0.0
offset_12-repair	no_image	20.6	20.6	0.0
nearest_id-repair	no_image	19.9	19.9	0.0
degree-repair	no_image	17.0	17.0	0.0
min_repair-repair	no_image	17.0	17.0	0.0
max_repair-repair	no_image	39.2	39.2	0.0
seeded-repair	no_image	24.1	24.1	0.0
card_first-repair	multimodal	23.6	23.6	23.6
card_first-atomic	atomic_binding	23.6	–	23.6
silhouette-repair	multimodal	78.3	78.3	78.3
silhouette-atomic	atomic_binding	78.3	–	78.3
appearance-repair	multimodal	54.6	54.6	54.6
appearance-atomic	atomic_binding	54.6	–	54.6
oracle-repair	oracle_binding	100.0	100.0	100.0

graph. Reviewers must be able to identify the reference, read candidate IDs, distinguish the unique match and understand the task text. Ambiguous or contradictory items are not repaired by changing the gold to match a vote. They require stimulus revision and new qualification with versioned lineage.

Six reviewer roles are assigned per construction, two per visual arm. Repeated fault packets and ID replicates within that arm may be inspected by those same reviewers; they are explicitly dependent reviews, not twelve independent judgments. No reviewer receives both the anchor and a different visual arm of the construction. Adjudication must be independent of the two original reviewers. No genuine reviewer records are present in this release.

The construction eligibility artifact binds the study and queue locks, raw qualification receipts, adjudications and recomputed decisions. Validation reruns the qualification rules instead of trusting an editable list of retained IDs. A rejected arm rejects its construction; otherwise incomplete reviews remain pending and unresolved concerns require adjudication. Only fully qualified constructions enter the common qualified subset. All twelve observations of a retained construction enter together under every condition. The full set remains separately available. An empty qualified set currently reflects pending collection.

The local study manifest fixes input/evaluator versions, conditions, baseline configurations and planned receipt slots. External models require a separate locked provider revision, adapter, precision, image policy, output limits and failure policy. The historical 15-model inventory is a planning list, not a claim of supported live endpoint availability. Additional thinking/effort modes are not claimed without an executable pinned configuration. Neither an empty result ta-

ble nor a synthetic fixture is accepted as an empirical receipt. The external lock also embeds validated construction eligibility before collection and retains all full-set observation slots. Analysis cannot substitute another eligibility artifact after outcomes are known. Model payloads exclude qualification evidence. Nonzero adapter exits and timeouts preserve stdout, stderr, exit status and available partial streams, with exact bytes retained alongside readable text. They remain failed attempts without retry.

6. Planned Learned-Model Matrices

The main paper retains the complete model inventory and primary multimodal matrix. Tables 5 and 7 reserve the decomposed conditions; Table 6 reserves the joint and paired readout. Models compare identical construction sets and exact native inputs. Percentages, eligible group counts, paired differences and missing/invalid outcomes must be reported together.

The table schema has 212 null-valued result fields. They render as genuine empty cells, not zeroes or placeholders that could be mistaken for scores. No old atomic result is transferred into a new primary input.

7. Source Context and Reproducibility

The full 24-source gallery is provided for resource inspection (Figure 1). Source scale bands D1–D4 describe assembly size, not cognitive task difficulty. Original part transforms remain preserved. Isolated reference/candidate cards are views of source parts, not new mechanically feasible assemblies. Graph faults are abstract unavailable vertices, not physical extraction animations.

All figure crops, native image hashes, graph inputs, GT

Table 5. Planned VLM condition decomposition, with all results blank. Atomic is binding-only accuracy; No-image and Oracle are exact repair under their respective information conditions. Repair|Bind is multimodal repair conditional on correct binding. Invalid is the multimodal invalid-output rate.

Model	Atomic	No-image	Oracle	Repair Bind	Invalid
GPT-6 Astra					
GPT-5.6 Sol					
GPT-5.6 Luna					
Claude Fable 5.1					
Claude Sonnet 5					
Gemini 3.1 Pro Preview					
Gemini 3.8 Flash					
Qwen3.5-9B					
Qwen3.5-27B					
InternVL3.5-8B					
InternVL3.5-38B					
Gemma 3 4B IT					
Gemma 3 27B IT					

Table 6. Planned task-aligned interface readout; all entries are unmeasured. J is the equal-group joint-condition failure percentage; n_{AO} and G_{AO} count jointly competent observations and groups. ΔRepair is multimodal minus oracle exact repair, and ΔBind is multimodal minus atomic binding, in equal-group percentage points. Full and qualified versions, micro estimates, all eight contingency counts and shared-group sensitivity intervals accompany the complete machine-readable analysis. Empty eligibility remains undefined.

Model	J	n_{AO}	G_{AO}	ΔRepair	ΔBind
GPT-6 Astra					
GPT-5.6 Sol					
GPT-5.6 Luna					
Claude Fable 5.1					
Claude Sonnet 5					
Gemini 3.1 Pro Preview					
Gemini 3.8 Flash					
Qwen3.5-9B					
Qwen3.5-27B					
InternVL3.5-8B					
InternVL3.5-38B					
Gemma 3 4B IT					
Gemma 3 27B IT					

Table 7. Planned text-only model conditions; no image-bearing score is imputed. Empty cells are unmeasured values.

Model	No-image repair	Oracle repair
Qwen3-32B		
DeepSeek-R1-Distill-Qwen-32B		

and source links are stored in the publication provenance manifest. The paper generator reads counts, representative examples and algorithm outcomes from the current version’s generated records. Rebuilding a figure does not collect a model response. The main PDF validator enforces at most eight content pages before references, unmodified source CVPR style and resolved citations.

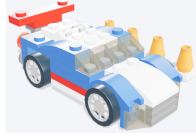
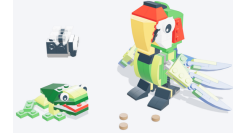
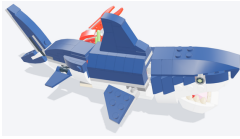
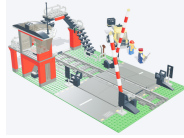
D1 | 31028 | 53 parts
Sea Plane**D1 | 31027 | 67 parts**
Blue Racer**D1 | 10156 | 111 parts**
LEGO Truck**D1 | 42102 | 129 parts**
Mini CLAAS XERION**D1 | 42020 | 141 parts**
Twin-Rotor Helicopter**D2 | 10036 | 166 parts**
Pizza To Go**D2 | 10014 | 170 parts**
Caboose**D2 | 31031 | 215 parts**
Rainforest Animals**D2 | 31088 | 230 parts**
Deep Sea Creatures**D2 | 42004 | 246 parts**
Mini Backhoe Loader**D2 | 42061 | 261 parts**
Telehandler**D2 | 31009 | 270 parts**
Small Cottage**D2 | 5867 | 278 parts**
Super Speedster**D2 | 10128 | 336 parts**
Train Level Crossing**D3 | 10159 | 592 parts**
City Airport**D3 | 10001 | 848 parts**
Metroliner**D4 | 10269 | 1,029 parts**
Harley-Davidson Fat Boy**D4 | 10242 | 1,086 parts**
MINI Cooper**D4 | 42066 | 1,244 parts**
Air Race Jet**D4 | 10213 | 1,254 parts**
Shuttle Adventure**D4 | 10265 | 1,357 parts**
Ford Mustang**D4 | 10220 | 1,428 parts**
Volkswagen T1 Camper Van**D4 | 21309 | 1,845 parts**
NASA Apollo Saturn V**D4 | 42054 | 1,978 parts**
CLAAS XERION 5000 TRAC VC

Figure 1. The 24 original source assemblies. Images preserve actual geometry and source poses. This source inventory is distinct from the subset of admitted repair constructions and its dependence groups. All assemblies retain source attribution in the release.